

Sequence and Series

10.1, 10.2 → Sequence & its Convergence
 10.3 → Series

$f: \mathbb{R} \rightarrow \mathbb{R}$ $y = f(x)$ is called
 real valued function. $\rightarrow f: (\mathbb{N}) \rightarrow \mathbb{R}$

Sequence:

Sequence is a function with domain as set of natural numbers/whole numbers.

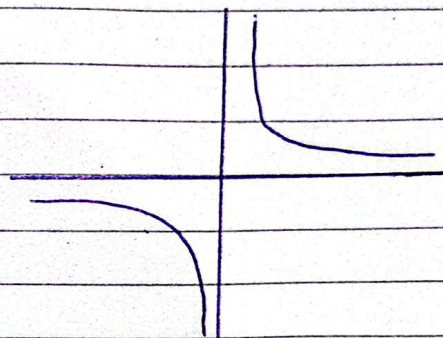
$$f: \mathbb{N} \rightarrow \mathbb{R}$$

$$y = f(n) = a_n \text{ } n\text{th term}$$

$$\{a_n\}$$

$$f(x) = \frac{1}{x}$$

$$f(n) = \frac{1}{n} = a_n$$



$$a_n = \frac{1}{n}, \quad n = 1, 2, 3, \dots$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

$$\left\{ \frac{1}{n} \right\}$$

- Limit of the function is always unique

Limit of sequence:

$$\lim_{n \rightarrow \infty} a_n \rightarrow L \text{ (finite)}$$

then sequence is convergent
otherwise it is divergent.

i) $\left\{ \frac{1}{n} \right\}$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \rightarrow 0$$

$\Rightarrow \left\{ \frac{1}{n} \right\}$ is convergent.

ii) $\left\{ \frac{n}{n+1} \right\}$

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} \left(\frac{\infty}{\infty} \right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1 \text{ (finite)}$$

$\Rightarrow \left\{ \frac{n}{n+1} \right\}$ is convergent.

• $\left\{ (-1)^n \right\} \quad \left\{ -1, 1, -1, 1, \dots \right\}$

$$a_n = (-1)^n$$

$$\lim_{n \rightarrow \infty} (-1)^n \begin{cases} \rightarrow 1 \rightarrow n \text{ even} \\ \rightarrow -1 \rightarrow n \text{ odd} \end{cases}$$

Limit is not unique. So sequence is not convergent.

$$i) \lim_{n \rightarrow \infty} \frac{\ln n}{n} \rightarrow 0$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1} \rightarrow 0$$

$$ii) \lim_{n \rightarrow \infty} \frac{n \sqrt[n]{n}}{1}$$

$$\lim_{n \rightarrow \infty} n^{1/n} \rightarrow 1$$

$$iii) \lim_{n \rightarrow \infty} x^{1/n} \rightarrow 1 \quad x > 0$$

$$\lim_{n \rightarrow \infty} 2^{1/n} \rightarrow 1$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^{1/n} \rightarrow 1$$

$$iv) \lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1) \\ (-1, 1)$$

$$x = -\frac{1}{2}$$

$$\lim_{n \rightarrow \infty} \left(-\frac{1}{2}\right)^n \rightarrow 0$$

$$a_n = \left(-\frac{1}{2}\right)^n \quad \left\{ -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, \dots \right\}$$

$$v) \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

Example:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{2}{n}\right)^n = e^2$$

$$vi) \lim_{n \rightarrow \infty} \frac{x^n}{n!} \rightarrow 0 \text{ for any } x.$$

Example:

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} \rightarrow 0$$

$$2, \frac{4}{2}, \frac{8}{6}, \frac{16}{24}, \dots$$

Series:

Given a sequence $\{a_n\} = \{a_1, a_2, a_3, \dots\}$
the expression of the form

$a_1 + a_2 + a_3 + a_4 + a_5 + \dots + a_n$
is called series of numbers with
infinite term.

$$\sum_{j=1}^{\infty} a_j = \sum_{n=1}^{\infty} a_n \rightarrow n\text{-th term}$$

Summation dummy index

Convergence of Series

There are several
Convergence Test
like (Integral Test)
 $\int_1^{\infty} a_n \, dn$ | $\int_1^{\infty} f(x) \, dx$

Sum of Series
(Always Not Possible)

Convergen Test

Series like geometric series whose sum is possible.

$$\left\{ \left(\frac{1}{3} \right)^n \right\} = \left\{ 1, \frac{1}{3}, \frac{1}{3^2}, \frac{1}{3^3}, \dots \right\} \text{ Geometric Sequence}$$

$$1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \quad \text{Geometric Series.}$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{3} \right)^n = \sum_{j=0}^{\infty} \left(\frac{1}{3} \right)^j = \sum_{l=0}^{\infty} \left(\frac{1}{3} \right)^l$$

Dummy Indices.

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{Sum of } n\text{-terms of } G\text{-series}$$

$$\lim_{n \rightarrow \infty} S_n = \frac{a}{1-r} \quad |r| < 1$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{3} \right)^n \quad G\text{-Series with } r = \frac{1}{3} < 1$$

$$= \frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{1}{2/3} = \frac{3}{2} \text{ Convergent.}$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$

Sum is not possible

$$f(x) = \frac{1}{x} \quad (1, \infty)$$

Integral Test

$$= \int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{x} dx$$

$$= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx = \lim_{b \rightarrow \infty} \ln(x) \Big|_1^b$$

$$= \ln(b) - \ln(1) = \ln(\infty) = \infty$$

Divergent Series and Sequence.